

Data Science

BSc Computer Science - Durham University

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1 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

1.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 1. Additional formatives are also shown in *italics*.

Title	Opening	Due	Feedback	Professor	Weight
Data Cleaning & Analytics	Week 11	-	11th Feb	Dr Igor Razgon	25%
Image Processing	15th Jan	2nd Mar	6th May	Dr George Koulieris	25%
Probability and Graphics	May Exam	-	-	Razgon & Koulieris	50%

Table 1: Assessments for Data Science.

Note: The Data Cleaning & Analytics test in Week 11 is closed book, and will be held during practicals (mine is Friday 4pm).

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	-

Table 2: Michaelmas Term Timetable for Data Science.

2 Probability and Statistics

This section is taught by Dr Igor Razgon. The recommended reading for this topic is:

- Probability and Statistics (Degroot, M.H. and Schervish, M.J., 2012).

2.1 Probability and Statistics: Basic Concepts

2.1.1 Counting Theory

Definition 2.1.1.1: The product rule states that, if a procedure can be broken down into a sequence of two tasks with n_1 ways to do the first task and n_2 ways to do the second task, then there are $n_1 n_2$ ways to complete the entire procedure.

Example 2.1.1.1: John wants to fly to Canada from Germany with a connecting flight through the UK. There are 3 flights from Germany to the UK, and 2 flights from the UK to Canada. How many possible flight combinations are there?

Solution: $3 \times 2 = 6$.

□

Definition 2.1.1.2: The sum rule states that, if you consider two independent sub-tasks (we only have to do one), if the first can be done in n_1 ways, and the second can be done in n_2 ways, then the total number of ways to complete the task is $n_1 + n_2$.

Example 2.1.1.2: Mary wants to choose a student project from three lists: lecturer 1 offers 5 projects, lecturer 2 offers 2 projects, and lecturer 3 offers 10 projects. How many possible options for her?

Solution: $5 + 2 + 10 = 17$.

□

Definition 2.1.1.3: The inclusion-exclusion principle states that, if two tasks can be done at the same time (unlike the sum rule in which they are independent), then there is a possibility of overcounting since some ways may be counted twice. To counter this, we add up the number of ways to do each task, then subtract the number of ways to do both tasks.

$$|A \cup B| = (|A| + |B|) - (|A \cap B|) \quad (1)$$

$$|A \cup B \cup C| = (|A| + |B| + |C|) - (|A \cap B| + |A \cap C| + |B \cap C|) + (|A \cap B \cap C|) \quad (2)$$

Definition 2.1.1.4: An r-permutation of a set of r distinct objects is one ordered arrangement. We denote the number of r -permutations of a set with n elements in as $P(n, r)$.

$$P(n, r) = n \times (n - 1) \times (n - 2) \times \dots \times (n - r + 1) = \frac{n!}{(n - r)!} \quad (3)$$

Suppose that a set consists of n objects, of which n_1 are of one type (indistinguishable from one another), n_2 are of a second type, ..., and n_k are of a k_{th} type where $n = n_1 + n_2 + \dots + n_k$. The number of permutations of the objects can be calculated with (4).

$$P(n; n_1, n_2, \dots, n_k) = \frac{n!}{n_1! n_2! \dots n_k!} \quad (4)$$

Definition 2.1.1.5: A r-combination is a counting of distinct objects in an unordered arrangement of length r . The number of r combinations of a set with n distinct elements is given by (5).

$$C(n, r) = \binom{n}{r} = \frac{n!}{(n - r)! \times r!} \quad (5)$$

Example 2.1.1.3: Assume elements A_1, A_2, A_3, A_4 are in a set. We want to choose 2 elements from them.

Solution:

- If the order of elements is important: $P(4, 2) = \frac{4!}{(4-2)!} = 12$.
- If the order of elements is not important: $C(4, 2) = \frac{4!}{(4-2)!2!} = 6$.

□

2.1.2 Probability Theory

Definition 2.1.2.1: Probability is the formalisation of uncertain events. It enables one to infer probabilities of interest based on assumptions and observations.

Definition 2.1.2.2: Statistics is the practice of the science of collecting and analysing data. It deals with experimental data and extracting knowledge from it.

Definition 2.1.2.3: An experiment is a procedure that yields one of a given set of possible outcomes.

Definition 2.1.2.4: The sample space of the experiment is the set of possible outcomes.

Definition 2.1.2.5: An event is any subset of the sample space.

Theorem 2.1.2.1: The Theory of Probability states that, given a finite sample space of all outcomes S , assuming that all outcomes are equally likely, for an event set E where $E \subseteq S$, then the probability of occurrence of the event is given by (6).

$$P(E) = \frac{|E|}{|S|} \quad (6)$$

Proof: Proof omitted. □

Definition 2.1.2.6: The axioms of probability are as follows:

- $P(E)$ is always between 0 and 1: $0 \leq P(E) \leq 1$.
- $P(S) = 1$.
- For disjoint events $E_1, E_2, \dots, E_n$ (with no intersection), $P(E_1 \cup E_2 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i)$.

Then, from the Axioms of Probability above, we can derive the following:

- **Definition 2.1.2.7:** The complement rule states that $P(\overline{E}) = P(S - E) = P(S) - P(E) = 1 - P(E)$.
- **Definition 2.1.2.8:** The impossible rule states that $P(\emptyset) = 0$.
- **Definition 2.1.2.9:** The general addition rule states that $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.
- If $E_1 \subseteq E_2$, then $P(E_1) \leq P(E_2)$.

Example 2.1.2.1: A sequence of 10 bits is randomly generated. What is the probability that at least 1 of these is 0?

Solution: Let E be an event in which all bits are 1.

$$P(E) = \frac{|E|}{|S|} = \frac{1}{2^{10}}$$

Then, by the complement rule, $P(\overline{E}) = P(1 - E)$ where $\overline{E}$ is the event in which at least one bit is not 1.

$$P(\overline{E}) = 1 - \frac{1}{1024} = \frac{1023}{1024}$$
□

2.2 Dependency and Bayes Theorem

2.3 Discrete Probability Distributions

2.4 Continuous Probability Distributions

2.5 Gaussian Distribution and its Characteristics

2.6 Correlations

2.7 Central Limit Theorem and Sampling

2.8 Hypothesis Testing

2.9 Data Types and Visualisation

2.10 Basic Data Analysis with Python

3 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon. [Check for recommended reading.](#)

4 Computer Graphics

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

5 Image Processing

This section is taught by Dr George Alex Koulieris. [Check for recommended reading.](#)

Part I

Appendix

6 Bibliography

6.1 Module Professors

Module Lead: Professor Thomas Erlebach

- Focus: Algorithms and networks.
- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00.

Lecturer: Doctor George Alex Koulieris

- Focus: Image processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

6.2 References

6.3 List of Figures

6.4 Reading List

6.4.1 Machine Learning

Degroot, M.H. and Schervish, M.J. (2012). Probability and Statistics. Boston Mass.: Pearson.